

Feature Scaling and Comparative Analysis of Fitness Dataset using StandardScaler and Min-Max Scaler

By Neha Rose Biju

Dataset Description

This is a fitness related dataset that contains the columns

Steps_Per_Day Calories_Burned Workout_Minutes

Since the features have different ranges we apply scaling .

```
# Dataset information
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   User_ID               100 non-null   object
1   Steps_Per_Day         100 non-null   int64
2   Calories_Burned       100 non-null   int64
3   Workout_Minutes       100 non-null   int64
dtypes: int64(3), object(1)
memory usage: 3.3+ KB
None
```

Feature scaling is used to standardize the data. It is because raw data can have different units and ranges in them. This can negatively impact our interpretations while using machine learning techniques. Two of the most commonly used scaling techniques are Standard scaler and Minmax scaler.

StandardScaler

$$z = \frac{x - \mu}{\sigma}$$

x = original value
 μ = mean
 σ = standard deviation

It follows standard normal distribution and transforms the data so that the mean become 0 and the standard deviation becomes 1. The mean of the data is subtracted and then divided by standard deviation.

Python Code:

```
# Standard Scaling
standard_scaler = StandardScaler()
X_standard = standard_scaler.fit_transform(X)
standard_df = pd.DataFrame(
    X_standard,
    columns=features
)
print("Standard Scaled Data")
standard_df.head()
```

Standard Scaled Data

	Steps_Per_Day	Calories_Burned	Workout_Minutes
0	-0.159517	0.183726	0.678568
1	1.722954	0.920927	1.543380
2	-1.574958	-0.965159	-1.607006
3	-0.574654	-0.773678	1.481608
4	1.198071	-0.448161	0.462365

Here the data is converted so that the mean becomes closer to 0 and the standard deviation becomes closer to 1. It is mostly used in linear regression, SVM, K-means clustering etc.

Min-Max Scaler

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

```

# Min-Max Scaling
minmax_scaler = MinMaxScaler()
X_minmax = minmax_scaler.fit_transform(X)
minmax_df = pd.DataFrame(
    X_minmax,
    columns=features
)
print("Min-Max Scaled Data")
minmax_df.head()

```

Min-Max Scaled Data

	Steps_Per_Day	Calories_Burned	Workout_Minutes
0	0.450130	0.555245	0.730769
1	0.992054	0.770629	1.000000
2	0.042655	0.219580	0.019231
3	0.330621	0.275524	0.980769
4	0.840951	0.370629	0.663462

The data is converted so that all the values lie between 0 to 1. This ensures that we do not have to deal with any negative values and it preserves the original distribution shape. It is most commonly use in KNN ,deep learning models ,neural networks etc.

Differences

In Standard Scaler the range is not fixed and the mean is always clod=ser to 0 and the standard deviation becomes 1. There is a possibility of negative values and the distribution is standardized.

In minmax scaler the range is fixed to between 0 to 1 and there is no fixed mean or standard deviation. There is no possibility for negative values and the distribution is normalized.

Results and Observations

Scaling is essential so that accurate observations can be obtained from the data. Every technique has its own advantages and disadvantages and is selected according to the requirements to satisfy. The choice mainly depends on the machine learning algorithm and data characteristics.